

Caitlin Chiang

Singapore, Singapore

🌐 <https://caitlinchiang.xyz/>

in <https://linkedin.com/in/caitlin-chiang>

8451-0986

chiangcaitlin2003@gmail.com

Summary

Innovative AI Software Engineer with extensive experience in architecting and deploying production-ready AI platforms, multi-agent systems, and scalable web applications. Proven track record in translating complex business and clinical requirements into high-impact technical solutions, ranging from healthcare analytics dashboards to generative AI-powered engagement platforms. Expert in full-stack development, LLM integration, and agentic workflows, with a strong foundation in system architecture and data engineering. Adept at leading cross-functional teams and mentoring engineers to deliver robust, scalable, and user-centric software in fast-paced, mission-driven environments.

Work Experience

- Liminal** Singapore, Singapore
AI Software Engineer Now - Now
 - Engineered a production-ready full-stack multi-agent AI platform enabling configurable workflows, dynamic query execution, and customizable result management.
 - Developed a responsive React-based chat interface featuring editable tables, configurable schemas, and real-time data visualization to enhance user productivity.
 - Architected a modular backend platform integrating multiple LLM providers, sophisticated prompt engineering strategies, and extensible AI workflows for scalable feature development.
- Xweave.io** Singapore, Singapore
Software Engineer May 2025 - Now
 - Architected and delivered secure integrations with multiple digital asset exchange providers, enabling scalable cross-border payment orchestration through standardized REST APIs.
 - Designed and productionized an event-driven webhook architecture to provide reliable, real-time transaction synchronization across distributed payment systems.
 - Architected and implemented a modular mid-market rate service using object-oriented design principles for efficient retrieval and synchronization of fiat and cryptocurrency rates.
 - Led and mentored two software engineers through technical guidance, code reviews, and architectural discussions to improve delivery velocity.
- Moya** Singapore, Singapore
Software Engineer Oct 2024 - Mar 2025
 - Engineered automated data processing pipelines to aggregate, transform, and consolidate large CSV datasets into standardized analytical outputs and master datasets.
 - Designed reusable scripts to generate abatement curves and analytical datasets, significantly improving data processing efficiency and consistency.
- SUTD campusX** Singapore, Singapore
Software Engineer Jan 2023 - Dec 2023
 - Collaborated with university faculty to architect and deliver an AI-powered learning platform enhancing educational workflows and student engagement.

- Engineered a modular application framework configurable for multiple university courses.
- Defined the application’s system architecture and relational database design, prioritizing scalability and long-term extensibility.

- **Jublia** Singapore, Singapore
Software Engineer *Aug 2023 - Dec 2023*

- Redesigned a user-centric dashboard experience, significantly improving usability and application navigation.
- Modernized a legacy Vue.js application by refactoring reusable components into a modular, maintainable architecture.
- Automated database integrity validation through scheduled monitoring scripts, reducing manual verification efforts and improving reliability.

- **Dashlabs.ai** Singapore, Singapore
Software Engineer *Dec 2020 - Dec 2021*

- Led and mentored a team of seven software engineers, providing technical leadership and architectural guidance for healthcare analytics solutions.
- Collaborated with medical professionals to translate complex clinical requirements into scalable analytical software.
- Delivered over 30 production features that streamlined healthcare workflows and improved operational efficiency.
- Engineered over 20 interactive analytics dashboards and architected data aggregation pipelines to transform complex healthcare datasets into actionable insights.

- **Dragonpay Corporation** Singapore, Singapore
Freelance Software Engineer *Jun 2021 - Jul 2021*

- Developed and deployed an internal employee wellness platform to streamline organizational health tracking and reporting.
- Engineered an internal expense management application that automated financial workflows and improved operational efficiency for the finance team.

- **Self-Employed** Singapore, Singapore
Freelance Software Engineer *Mar 2020 - Jun 2021*

- Partnered with business owners to translate strategic objectives into scalable web application solutions.
- Designed and developed end-to-end e-commerce platforms spanning customer-facing experiences and administrative management systems.
- Architected application databases and backend systems with a focus on maintainability and long-term growth.

- **Philippine Red Cross** Singapore, Singapore
Software Engineer *Jul 2021 - Oct 2021*

- Developed automated module-level testing scripts to improve software reliability, maintainability, and deployment confidence.

Projects

- **AI-Powered Public Engagement Platform**
Team Lead and Software Engineer *Sept 2024 - Dec 2024*
 - Led a team of four engineers to architect an AI platform that transforms user ideas into AI-generated visual concepts.

- Integrated advanced generative AI techniques, including prompt engineering and image inpainting, to improve output quality.
- Designed a modular architecture enabling administrators to customize engagement workflows without code changes.
- Ensured strict consistency between UX design specifications and production-ready software implementation.

• **Consumer Analytics & Data Visualization Platform**

Technical Lead and Software Engineer

May 2023 - Aug 2023

- Led a team of six engineers to deliver a data visualization platform for P&G's R&D organization to analyze consumer behavior.
- Engineered automated CSV ingestion and processing pipelines to transform raw datasets into interactive analytical visualizations.
- Developed interactive dashboards enabling researchers to compare multi-dimensional datasets via configurable timelines and charts.
- Integrated AI-powered image analysis to extract visual attributes and emotional signals from consumer-submitted images.

• **Queue Simulation & Operational Planning Dashboard**

Team Lead and Software Engineer

Jan 2024 - May 2024

- Led a team of five engineers to develop a configurable web-based queue simulation platform for banking branch operational planning.
- Architected a scalable and modular system design to support flexible simulation scenarios and future platform extensibility.

Education

• **Singapore University of Technology and Design (SUTD)**

Computer Science and Digital Humanities

Singapore, Singapore

Sept 2021 - May 2025

Technical Skills

Python, TypeScript, JavaScript, Ruby, React.js, Next.js, Vue.js, Node.js, Express.js, Flask, Ruby on Rails, LLM, LangChain, LangGraph, Prompt Engineering, AI Workflows, Multi-Agent Systems, Full-Stack Development, REST APIs, GraphQL (Apollo), MySQL, PostgreSQL, MongoDB, Firebase, AWS, GCP, Netlify, System Architecture, Relational Databases, Database Architecture, Data Pipelines, ETL, Data Visualization, UI/UX Design, Automated Testing, Cypress, Git, HTML, CSS, Material UI, Chakra UI, Cloudinary